Universität Freiburg – Institut für Physik

Applied Theoretical Physics
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Computational Physics: Materials Science - Exercise 8, SS 2026

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In this exercise, we will implement a function which measures the pressure of the system. Afterwards, we compare our pressure to the theoretical pressure of the system, dependent on the density, by calculating the equation of state.

Enrich your existing Lennard-Jones MD simulation from sheet 6 by implementing a function which can calculate the pressure of the system. We will calculate the pressure with equation (1)

$$P = \frac{1}{3V}(2K + \vartheta), \tag{1}$$

where P is the instantenous pressure of the system, V is its (constant) volume, K is the kinetic energy and ϑ is the so-called *virial*. You can calculate the virial via equation (2)

$$\vartheta = \sum_i^N \mathbf{r}_i \cdot \mathbf{f}_i = \sum_i^N \sum_{j>i} \mathbf{r}_{ij} \cdot \mathbf{f}_{ij}, \tag{2}$$

where $\mathbf{f}_i$ is the total force acting on the i-th atom and $\mathbf{f}_{ij}$ is the pair force due to the pair potential. Both formulations of the virial are equivalent. **Note that you still need to respect minimum-image convention**.

Problem 8. Equation of state of a Lennard Jones gas

Our simulations should be efficient at this point (make use of **numba**), so use at least N=125 particles at least and equilibrate your simulation for n_{eq} steps before starting your production run with n_{prod} steps. Use the parameters from table 1. Make sure that your macroscopic quantities are well converged.

(a) Calculate the equation of state. An equation of state relates the pressure P of the system to its density ρ. We will use equation (3)

$$\beta P/\rho \approx 1 + B_2\rho, \tag{3}$$

where B_2 is the second virial coefficient and $\beta = 1/k_BT$ with Boltzmann constant k_B as an approximation to the equation of state. You can calculate B_2 as an integral over the Mayer-f function, see below:

$$B_2 = -\frac{1}{2}\int \mathrm{d}^3\mathrm{r}(\exp(-\beta U(r)) - 1). \tag{4}$$

Make sure to understand where this expansion and expression is coming from. Calculate and plot the coefficient B_2/σ^3 as a function of ε/k_BT. Interpret what happens when B_2 changes sign. What scaling with ρ would you expect for our system parameters in your simulations? For what densities is this equation valid?

(b) Implement a function to calculate the pressure of your system. Calculate the average pressure $\langle P \rangle_t$ in bar and report a statistical error on this value via block averaging.

(c) Compare the pressure of your system to the pressure predicted by equation (3) by repeating your simulation at different densities starting with $\rho = 0.05\ \sigma^{-3}$ and going up to $\rho = 0.50\ \sigma^{-3}$. What do you see?

What to submit
Hand in a zip file on ILIAS containing the following files:

- your Python source code
- a short report in PDF format with your main results

A good submission contains:

- a brief summary of the code base so far with an explanation on what you added to your simulation.
- plots which show your results in a clever way and demonstrate the expected scaling laws. If points do not follow theory, make reasonable effort to explain why.
- a snapshot from ovito to show that your simulation works.

How to prepare for the exercise group

- Make sure you can explain your results and the code.
- Keep your chat history with an AI chatbot to explain in the tutorial how you **interacted** with it to develop your code.
- Prepare yourself for the questions outlined below.

Questions to think about

Make sure to feel comfortable talking about your code, your results and the questions below. These questions provide the key take-away points of this exercise:

- Explain how you would implement a barostat and its implications. What kind of (more sophisticated) barostats do you know and what can you do with them?
- What does the sign of B_2 tell us about the microscopic intermolecular interactions?
- Why do we use block averaging in this case to calculate the average pressure?

Parameter	**Value**
N	125
n_{eq}	30000 steps
n_{prod}	30000 steps
m	39.95 g/mol
T	300 K
ε	$0.3\,k_B T$
σ	0.188 nm
Δt	2 fs
ρ	0.05 σ^{-3}

Table 1: Simulation parameters for this exercise sheet.